# Ayman Marzuq Zaman

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## Education

**University of Toronto** — **Toronto, Canada**
*Bachelor of Applied Science (BASc), Mechanical Engineering + PEY Co-op* — *April 2028 (expected)*
Minors in Robotics & Mechatronics | Engineering Business (Rotman) | Dean's List Scholar
Relevant courses: Mechanical Design Engineering, Manufacturing Engineering, Materials Science, Solid Mechanics, Fluid Mechanics, Heat Transfer, Thermodynamics, Python Programming, Circuits

## Skills

**CAD & Engineering Tools**: SolidWorks (CAD, Simulation, 3DExperience), ANSYS (FEA, CFD, Thermal Analysis), Engineering Drawings (GD&T, BOM creation)

**Programming & Analysis**: Python (NumPy, Pandas), C/C++, MATLAB (Simulink, Data Visualisation), MS Excel

**Machining & Fabrication**: Lathes, Mills, Saws, Waterjet Cutting, Sheet Metal, 3D Printing, Welding (exposure), Assembly

**Professional**: MS PowerPoint, MS Word, Project Management, Analytical Problem Solving, Cross-Team Communication

**Certifications**: CSWA (2024), WHMIS (2025), Basic Machining Course – George Brown College (2024)

## Work Experience

**MIE Machine Shop – MC78** — **University of Toronto**
*Machine Shop Assistant* — *May 2025–August 2025*

- Performed maintenance and calibration of CNC/manual lathes, mills, and saws in accordance with SOPs, ensuring precise alignment, lubrication, and tolerances were upheld to maintain quality and optimal machine performance.
- Conducted Root Cause Analysis & Corrective Action (RCCA) on mechanical and electrical faults using technical manuals and schematics, restoring machine functionality and reducing downtime by **30%**.
- Independently designed and prototyped a crane in SolidWorks using aluminum extrusions and McMaster-Carr components to lift and position heavy lathe chucks, reducing setup time by **80%** (∼30+ min per setup).
- Provided hands-on technical guidance to students on machining setup, tooling selection, and process optimization, ensuring adherence to safety protocols and quality standards.

## Project Experience

**UT Baja Racing Team** — **University of Toronto**
*Co-Lead, Brake Team* — *June 2025–Present*

- Leading a six-member team designing and validating the complete braking system for the first-ever UT Baja vehicle, maintaining compliance with Baja SAE ruleset (Article B.7) governing hydraulic circuits, pedal loads, and brake line specifications.
- Performed calculations for static structural, dynamic, torsional and thermal performance in Microsoft Excel, benchmarking against prior competition data and research literature to derive system specifications.
- Designing the brake pedal assembly in SolidWorks, applying DFM/DFA principles to translate geometric and load constraints derived from analytical data into manufacturable parts.
- Coordinating with Suspension, Drivetrain, Chassis, and Electrical subteams to align brake system interfaces with vehicle-level design and integration requirements.

**UT Formula Racing Team** — **University of Toronto**
*Team Member, Aerodynamics Team* — *August 2024–Present*

- Redesigned the rear wing mounting structure for the UT24 Formula SAE vehicle to support aerodynamic loads while achieving a **12%** weight reduction; vehicle placed **1st overall** at Formula Hybrid + Electric (NH).
- Developed and executed FEA test procedures in ANSYS to assess structural stiffness under static, vibration, and shock loading conditions; applied Topology Optimization to eliminate excess material and inform final geometry.
- Machined brackets, mounts, and accumulator walls using lathes, mills, and saws, accumulating **30+** hours of hands-on fabrication in support of vehicle assembly.

**Mechanical Engineering Design Course** — **University of Toronto**
*Team Leader, CNC Router Project* — *September 2024–December 2024*

- Led a four-member team through the full product design and development cycle of a compact 4'×2' CNC router in SolidWorks, delivering a design ∼**35%** less costly than comparable 4'×4' market alternatives.
- Integrated custom and McMaster-Carr components into a parametric CAD assembly, applying GD&T principles to define fits, surface finish, and assembly requirements for efficient fabrication.
- Compiled and presented system validation results and performance trade-off analyses to the Engineering Manager via MS PowerPoint, backed by market data, engineering calculations, and SolidWorks simulations.
- Prepared annotated 2D & 3D fabrication drawings, 30-part BOM, and technical documentation to support manufacturing and assembly; managed design changes through structured revision control.